JAGANNATH JANA

Address: Johann-Stelling Straße 29, Greifswald, 17489, Germany
Mobile No.+491639140954/ +919635588574 (Whatsapp)
Email: jagannath.jana88@gmail.com
Date of Birth: 10/03/1988
Nationality: Indian

SUMMARY

Accomplished structural biologist and biophysical chemist with expertise in nucleic acid dynamics and high-resolution NMR spectroscopy, specializing in G-quadruplex structures to support structure-guided drug design. Contributed multiple high-resolution PDB entries. Proficient in NMR and complementary biophysical techniques, with a comprehensive skill set for analyzing biomolecular stability and drug-target interactions, advancing therapeutic development through detailed molecular insights.

RESEARCH EXPERIENCE

Postdoctoral Researcher, University of Greifswald, Germany **06/2019 - 03/2024**
Supervisor: Professor Dr. Klaus Weisz

- Led research on nucleic acid structural dynamics, focusing on G-quadruplexes using high-resolution NMR spectroscopy.
- Characterized three-dimensional G-quadruplex structures, contributing high-resolution entries to the Protein Data Bank (PDB).
- Applied biophysical techniques, including Fluorescence Spectroscopy, UV-vis spectroscopy, Circular Dichroism (CD), ITC, and DSC, to analyze biomolecular interactions.

Postdoctoral Researcher, Institute Curie, France **01/2018 -02/2019**
Supervisor: Professor Dr. Stephan Vagner

- Researched 53BP1 protein interactions with nucleic acids to understand DNA damage response.
- Demonstrated 53BP1 binding using electrophoretic mobility shift assays and SPR.
- Characterized 53BP1 binding affinity and specificity via biophysical methods.

EDUCATION

Ph.D. at Bose Institute(Calcutta University, W.B., India) **2011 - 2017**
Thesis Title: *De novo* Designed Antimicrobial peptides and SmallMolecules: Potent Stabilizer of G-quadruplex Structures
Supervisor: Professor Dr. Subhrangsu Chatterjee

Master of Science in Chemistry **2009 - 2011**
Vidyasagar University, W.B., India

Bachelor of Science in Chemistry **2006- 2009**
Midnapore College (Vidyasagar University, W.B., India)

ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Bengali
- **Awards/Activities:** Qualified IIT Entrance Examination (JAM-2009/2010) (For pursuing M.Sc. in IIT), Merit Cum Means Scholarship from W.B. Govt. (2010-2011), CSIR Fellowship (2011) (For pursuing Ph.D.)

PUBLICATIONS

Total Citations = 802 **h-index** = 15 **i10-index** = 23

30. **Remodelling Ca2+ dynamics by targeting a promising E-box containing G-quadruplex at ORAI1 promoter in triple-negative breast cancer.**
Chatterjee, O.Ψ; **Jana, J.Ψ;** Panda, S.Ψ; Dutta, A.; Sharma, A.; Saurav,S.; Motiani, R.K.; Weisz, K.; Chatterjee, S.(Ψ**contributed equally**).
Cell Calcium, 2024, *123*, 102944 **(I.F. = 4.3)**

29. **A pH-Responsive Topological Switch Based on a DNA Quadruplex-Duplex Hybrid.**
Vianney,Y.M.; **Jana,J.;** Weisz, K.
Chemistry - A European Journal, 2024, *30 (29),* e202400722 **(I.F. = 4.3)**

28. **Impact of loop length and duplex extensions on the design of hybrid-type G-quadruplexes.**
Jana,J.; Vianney,Y.M.; Weisz, K.
Chemical Communications, 2024, *60(7),* 854-857. **(I.F. = 4.3)**

27. **53BP1 interacts with the RNA primer from Okazaki fragments to support their processing during unperturbed DNA replication.**
Leriche, M.Ψ; Bonnet, C.Ψ; **Jana, J.;** Chhetri, G.; Mennour, S.; Martineau, S.; Pennaneach,V.; Busso, D.; Veaute, X.; Bertrand, P.; Lambert, S.; Somyajit, K.; Uguen, P.; Vagner, S.
Cell Reports, 2023, *42(11)*, 113412. **(I.F. = 7.5)**

26. **Showcasing Different G-Quadruplex Folds of a G- Rich Sequence: Between Rule-Based Prediction and Butterfly Effect.**
Vianney, Y.M.Ψ; Schröder, N.Ψ; **Jana, J.Ψ**; Chojetzki, G.; Weisz, K.
Journal of the American Chemical Society, 2023, *145(40)*, 22194-22205.(Ψ**contributed equally**). **(I.F. = 14.4)**

25. **Guiding the folding of G-quadruplexes through loop residue interactions.**
Jana,J.; Vianney,Y.M.; Schröder,N.; Weisz, K.
Nucleic Acids Research, 2022, *50(12),* 7161-7175. **(I.F. = 16.6)**

24. **Prion Derived Tetrapeptide Stabilizes Thermolabile Insulin via Conformational Trapping.**
Mukherjee, M.; Das, D.; Sarkar, J.; Banerjee, N.; **Jana, J.**; Bhat, J.; Reddy, J.; Bharatam, J.; Chattopadhyay, S.; Chatterjee, S.; Chakrabarti, P.
iScience, 2021, *24(6)*, 102573. **(I.F. = 4.6)**

23. **Expanding the Topological Landscape by a G-Column Flip of a Parallel G-Quadruplex.**
Mohr, S.; **Jana, J.;** Vianney, Y.M.; Weisz, K.
Chemistry - A European Journal, 2021, *27(40)*, 10437-10447. **(I.F. = 4.3)**

22. **Thermodynamic Stability of G-Quadruplexes: Impact of Sequence and Environment.**
Jana, J.; Weisz, K.
ChemBioChem, 2021, *22(19)*, 2848-2856. **(I.F. = 2.6) (Top Downloaded Article)**

21. **Structural motifs and intramolecular interactions in non-canonical G-quadruplexes.**
Jana, J.; Mohr, S.; Vianney, Y.M.; Weisz, K.
RSC Chemical Biology, 2021, *2*, 338-353. **(I.F. = 4.2)**

20. **A Thermodynamic Perspective on Potential G-Quadruplex Structures as Silencer Elements in the MYC Promoter.**
Jana, J.; Weisz, K.
Chemistry - A European Journal, 2020, *26(71)*, 17242-17251. **(I.F. = 4.3) (Hot Paper)**

19. **Impact of a snap-back loop on stability and ligand binding to a parallel G- Quadruplex.**
Schnarr,L.; **Jana, J.;** Preckwinkel, P.; Weisz, K.
The Journal of Physical Chemistry B, 2020, *124(14)*, 2778-2787. **(I.F. = 2.8)**

18. **Enhancement of ABA sensitivity through conditional expression of ARF10 gene in Brassica juncea exhibit fertile plants with tolerance against Alternaria brassicicola.**
Mukherjee, A.; Mazumder, M., **Jana; J.**, Srivasatava, A.K.; Mondal, B.; De, A.; Ghosh, S.; Saha, U.; Bose, R.; Chatterjee, S.; Dey, N.; Basu, D.
Molecular Plant-Microbe Interactions, 2019, *32(10),* 1429-1447. **(I.F. = 3.5)**

17. **Nrf-2 transcriptionally activates P21Cip/WAF1 and promotes A549 cell survival against oxidative stress induced by H_2O_2.**
Jana, S.; Patra, K.; **Jana, J.,** Mandal, D.P. and Bhattacharjee, S.
Chemico-Biological Interactions, 2018, *285,* 59-68. **(I.F. = 4.7)**

16. **Spatial Position Regulates Power of Tryptophan: Discovery of Major Groove Specific Nuclear Localizing Cell Penetrating Tetrapeptide.**
Bhunia, D.; Mondal, P.; Das, G.; Saha, A.; Sengupta, P.; **Jana, J.;** Mohapatra, S.; chatterjee, S.; Ghosh, S.
Journal of the American Chemical Society, 2018, *140 (5),* 1697-1714**. (I.F. = 14.4)**

15. **LINCRNA00273 promotes cancer metastasis and its G-Quadruplex promoter can serve as a novel target to inhibit cancer invasiveness.**
Jana, S.$^{\Psi}$; **Jana, J.**$^{\Psi}$; Patra, K.; Mondal, S.; Bhat, J.; Sarkar, A.; Sengupta, P.; Biswas, A.; Mukherjee, M.; Tripathi, S.P.; Gangwal, R.; Sangamwar, A.T.; Mukherjee, G.; Bhattarjee, S.; Mandal, D.; Chatterjee, S.
Oncotarget, 2017, *8*, 110234. ($^{\Psi}$**contributed equally**). **(I.F. = 3.4)**

14. **A Small Molecule Impedes Insulin Fibrillation: Another New Role of Phenothiazine Derivatives.**
Mukherjee, M.; **Jana, J.**; Chatterjee, S.
Chemistry Open, **2017,** *7(1),* 68-79. **(I.F. = 2.5)**

13. **Restriction of telomerase capping by short non-toxic peptides *via* arresting telomeric G-quadruplex: a novel avenue in anti-cancer therapy.**
Jana, J.; Sengupta, P.; Mondal, S.; Chatterjee, S.
RSC Advances, **2017**, *7,* 20888-20899. **(I.F. = 3.9)**

12. **Chelerythrine down regulates expression of VEGFA, BCL2 and KRAS by arresting G-Quadruplex structures at their promoter regions.**
Jana, J.; Mondal, S.; Bhattacharya, P.; Saha, P.; Kundu, P.; Chatterjee, S.
Scientific Reports, **2017,** *7*, 40706. **(I.F. = 4.3)**

11. **Myricetin arrests human telomeric G-quadruplex structure: a new mechanistic approach as an anticancer agent.**
Mondal, S.; **Jana, J.**; Sengupta, P.; Jana, S.; Chatterjee, S.
Molecular Biosystems, **2016**, *12*, 2506-2518. **(I.F. = 3.0)**

10. **Functional characterization of Rorippa indica defensin and its efficacy against Lipaphis erysimi**.
Sarkar, P.; **Jana, J.**; Chatterjee, S.; Sikdar, S. R.
Springerplus, **2016,** *5*, 511. **(I.F. = 1.85)**

9. **Parkia javanica Extract Induces Apoptosis in S-180 Cells via the Intrinsic Pathway of Apoptosis.**
Patra, K.; Jana, S.; Sarkar, A.; Karmakar, S.; **Jana, J.**; Gupta, M.; Mukherjee, G.; De, U. C.; Mandal, D. P.; Bhattacharjee, S.
Nutrition and Cancer, **2016,** *68* (4), 689-707. **(I.F. = 2.9)**

8. **Reverse Watson-Crick G-G base pair in G-quadruplex formation.**
Mondal, S.; Bhat, J.; **Jana, J.**; Mukherjee, M.; Chatterjee, S.
Molecular Biosystems, **2016,** *12* (1), 18-22. **(I.F. = 3.0)**

7. **Plant alkaloid chelerythrine induced aggregation of human telomere sequence--a unique mode of association between a small molecule and a quadruplex.**
Ghosh, S.$^{\Psi}$; **Jana, J.**$^{\Psi}$; Kar, R. K.$^{\Psi}$; Chatterjee, S.; Dasgupta, D.
Biochemistry, **2015,** *54(4)*, 974-986. ($^{\Psi}$ **contributed equally)**. **(I.F. = 2.9)**

6. **Protective effect of antioxidant rich aqueous curry leaf (Murraya koenigii) extract against gastro-toxic effects of piroxicam in male Wistar rats.**
Firdaus, S. B.; Ghosh, D.; Chattyopadhyay, A.; Dutta, M.; Paul, S.; **Jana, J**.; Basu, A.; Bose, G.; Lahiri, H.; Banerjee, B.
Toxicology Reports, **2014,** *1*, 987-1003. **(I.F. = 3.95)**

5. **Antitumorigenic potential of linalool is accompanied by modulation of oxidative stress: an in vivo study in sarcoma-180 solid tumor model.**
Jana, S.; Patra, K.; Sarkar, S.; **Jana, J.**; Mukherjee, G.; Bhattacharjee, S.; Mandal, D. P.
Nutrition and Cancer, 2014, *66 (5)*, 835-848. **(I.F. = 2.9)**

4. **Indolicidin targets duplex DNA: structural and mechanistic insight through a combination of spectroscopy and microscopy.**
Ghosh, A.; Kar, R. K.; **Jana, J**.; Saha, A.; Jana, B.; Krishnamoorthy, J.; Kumar, D.; Ghosh, S.; Chatterjee, S.; Bhunia, A.
ChemMedChem, 2014, *9 (9),* 2052-2058. **(I.F. = 3.6)**

3. **Sequence context induced antimicrobial activity: insight into lipopolysaccharide permeabilization.**
Ghosh, A.[Ψ]; Datta, A.[Ψ]; **Jana, J**.[Ψ]; Kar, R. K.; Chatterjee, C.; Chatterjee, S.; Bhunia, A.
Molecular Biosystems, 2014, *10*(6), 1596-1612.([Ψ] **contributed equally**). **(I.F. = 3.0)**

2. **Human cathelicidin peptide LL37 binds telomeric G-quadruplex.**
Jana, J.; Kar, R. K.; Ghosh, A.; Biswas, A.; Ghosh, S.; Bhunia, A.; Chatterjee, S., **Molecular Biosystems, 2013,** *9 (7)*, 1833-1836. **(HOT Article as recommended by the referees). (I.F. = 3.0)**

1. **Novel G-quadruplex stabilizing agents: in-silico approach and dynamics.**
Kar, R. K.; Suryadevara, P.; **Jana, J.**; Bhunia, A.; Chatterjee, S**.**
Journal of Biomolecular Structure and Dynamics, 2013, *31* (12), 1497-1518. **(I.F. = 4.4)**

Book Chapter

Vianney, Y.M., **Jana, J.,** Schröder, N., Weisz, K. **Nucleic Acid Structure and Biology. In Nucleic Acid Biology and its Application in Human Diseases. Springer Nature, Singapore. (2023).**

Deposited NMR Structures in PDB

1. Jana, J., Vianney, Y.M., Schroeder, N., Weisz, K. **(2022).** Structure of a hybrid-type G-quadruplex with a snapback loop (hybrid 1R'). **PDB 7ZEK**. https://doi.org/10.2210/pdb7ZEK/pdb

2. Jana, J., Vianney, Y.M., Schroeder, N., Weisz, K. **(2022).** Structure of a parallel G-quadruplex with a snapback loop. **PDB 7ZEM**. https://doi.org/10.2210/pdb7ZEM/pdb

3. Jana, J., Vianney, Y.M., Schroeder, N., Weisz, K. **(2022).** Structure of a hybrid-type G-quadruplex with a snapback loop and an all-syn G-column (hybrid-1R). **PDB 7ZEO.** https://doi.org/10.2210/pdb7ZEO/pdb

4. Vianney, Y.M., Schroeder, N., Jana, J., Chojetzki, G., Weisz, K. **(2023).** Three-layered parallel G-quadruplex with snapback loop from a G-rich sequence with five G-runs. **PDB 8PSB**. https://doi.org/10.2210/pdb8PSB/pdb

5. Vianney, Y.M., Schroeder, N., Jana, J., Weisz, K. **(2023).** Three-layered basket-type G-quadruplex from a G-rich sequence with five G-runs. **PDB 8PSC.** https://doi.org/10.2210/pdb8PSC/pdb

6. Vianney, Y.M., Schroeder, N., Jana, J., Weisz, K. **(2023).** G-quadruplex with a 1-nt V-shaped loop from a G-rich sequence with five G-runs. **PDB 8PSI.** https://doi.org/10.2210/pdb8PSI/pdb

7. Jana, J., Vianney, Y.M., Schroeder, N., Weisz, K. **(2023).** (3+1) hybrid G-quadruplex from a G-rich sequence with five G-runs. **PDB 8PSE.** https://doi.org/10.2210/pdb8PSE/pdb

8. Jana, J., Vianney, Y.M., Weisz, K. **(2023).** Hybrid-1R G-quadruplex with a +(lpp) loop progression. **PDB 8R4E.** https://doi.org/10.2210/pdb8R4E/pdb

9. Jana, J., Vianney, Y.M., Weisz, K. **(2023).** (3+1) hybrid-2 G-quadruplex with a -(llp) loop progression. **PDB 8R4W.** https://doi.org/10.2210/pdb8R4W/pdb

10. Jana, J., Weisz, K. **(2024).** Solution structure of a parallel stranded G-quadruplex formed in ORAI1 promoter. **PDB 8RZX.** https://doi.org/10.2210/pdb8RZX/pdb.

11. Vianney, Y.M., Jana, J., Weisz, K. **(2024).** Structure of a quadruplex-duplex hybrid with a (-pd+l) loop progression. **PDB 8S1W.** https://doi.org/10.2210/pdb8S1W/pdb

SEMINARS & CONFERENCE

1. National seminar on Chemistry in the 21st Century for the Benevolence of the Society. (July 2011) at **Vidyasagar College, Kolkata**.
2. International Symposium (RTRC-2011) on Recent Trends of Research in Chemistry. (November 2011) at **Midnapore College, W.B.**
3. International Conference on Informatics & Integrative Biology (CIIB-2011). (December 2011) at **Bose Institute, Kolkata.**
4. Participated in the **5th Bruker Pre-NMRS Symposium 2013** at IIT Bombay on 3rd FEB.
5. Participated in the 19th **National Magnetic Resonance Society Meeting (NMRS-2013)** at **IIT Bombay** during 3-6th FEB, 2013.
6. 4th Indian Peptide Symposium (IPS 2013) at **SINP, Kolkata** during Feb 21-22, 2013.
7. Attended the '**Workshop on Multidimensional NMR and its Application**' held in **IISC, Bangalore** during March 7-9, 2013. (**selected out of 30 participants**)
8. Participated in the 21st **National Magnetic Resonance Society Meeting (NMRS-2015)** at **GNDU Amritsar** during 6-9th March, 2015.
9. Participated and presented poster in International Symposium on Chemical Biology and Drug Discovery (**ISCBDD-2016**) during 1-3rd March, 2016.
10. Attended workshop on **Molecular Docking, Virtual Screening & Biologics Discovery** hosted by Schrodinger and Bose Institute, Department of Biophysics, Kolkata, India on 5th and 6th December 2016.
11. Participated in conference on **G-QUADRUPLEXES: BENCHMARKING FROM STRUCTURES TO FUNCTIONS held in Institut Curie, Paris, France** on 20th February, 2018.

12. Attended workshop **G-NMR School 2020** hosted by **Max Planck Institute for Biophysical Chemistry**, taking place in **Göttingen, Germany** from February 17 – February 21, 2020.
13. Oral presentation in **8th International Meeting on Quadruplex Nucleic Acids, Marienbad, Czech Republic**, during 27 June 2022 – 1 July 2022.
14. Invited talk at the Department of Chemistry, Ashoka University, Delhi on 7th February, 2023

TECHNICAL SKILLS

Synthesis

Solid phase peptide synthesis

Biophysical Techniques

Fluorescence Spectroscopy, Absorbance Spectroscopy, Circular Dichroism Spectroscopy (CD), Isothermal Titration Calorimetry (ITC), Differential Scanning Calorimetry (DSC), Dynamic Light Scattering (DLS), Surface Plasmon Resonance (SPR)

Basic Molecular biology techniques

PCR, Native PAGE, GEL Electrophoresis, preliminary cell culture

NMR Spectroscopy

Biomolecular NMR structure calculation of peptide, protein and nucleic acid

Computational techniques

Homology Modelling of protein structures, Various Docking Softwares like, Zdock, and Schrodinger, MD simulation (AMBER), Pymol, VMD, Chimera.

REFERENCES

1. Prof. Dr. Subhrangsu Chatterjee, PhD
Department of Biophysics
Unified Academic Campus, Bose Institute
EN-80, Sector V, Bidhan Nagar
Kolkata - 700 091, India
E-mail: subhrangsu@gmail.com, subhro_c@jcbose.ac.in

2. Prof. Dr. Klaus Weisz, PhD
Institut für Biochemie
Universität Greifswald
Felix-Hausdorff-Str. 4
D-17487 Greifswald
Germany
Email: weisz@uni-greifswald.de

3. Prof. Dr. Stephan Vagner, PhD
Director CNRS UMR 3348 - INSERM U1278
Institut Curie-Centre de Recherche
Universite Paris Sud
F-91405 Orsay Cedex, France
Email: stephan.vagner@curie.fr